Credit Card Fraud

Classification Competition

Machine Learning Model Comparison & Hyperparameter Tuning

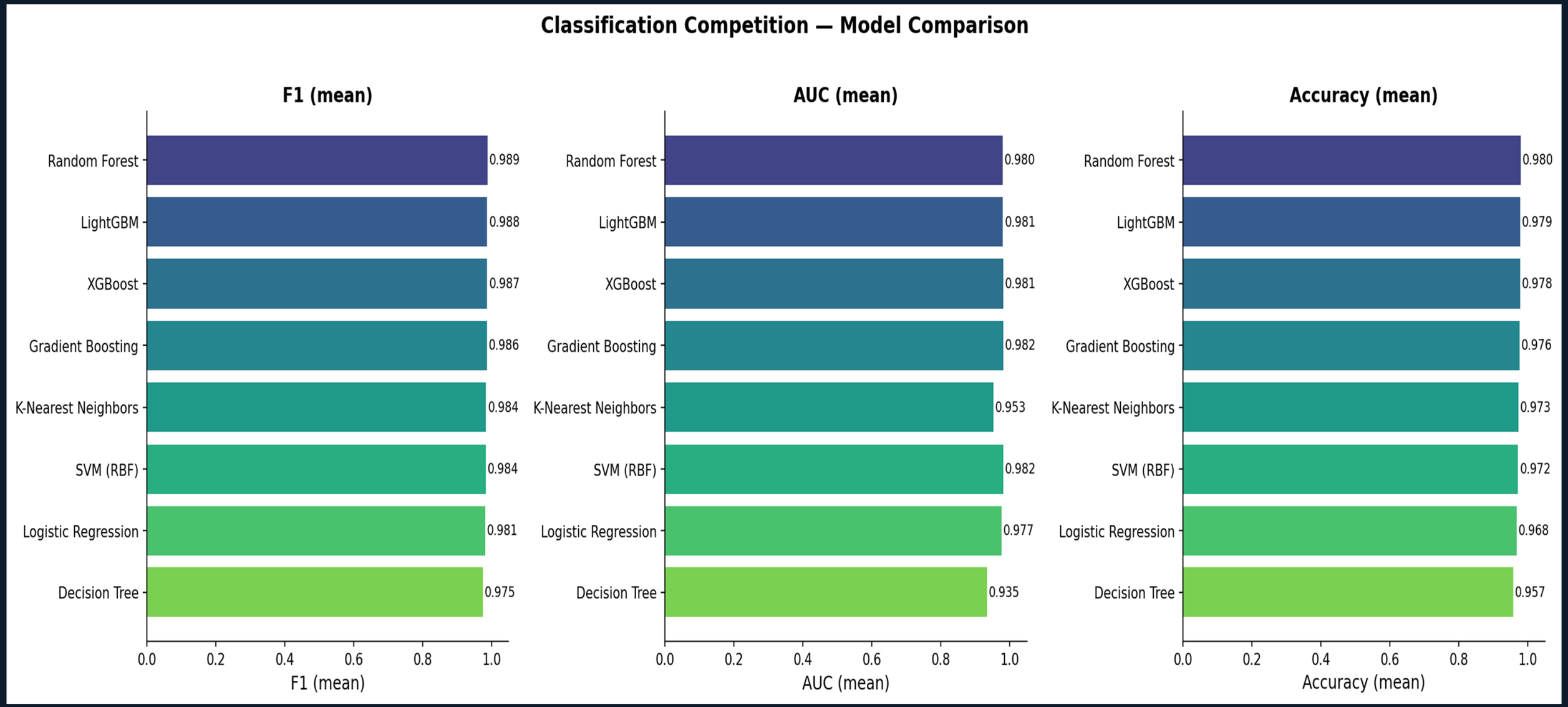
Dataset: 3,492 transactions | 29 features | Target: Class (Fraud / Not Fraud)

Fraud: 496 (14.2%) | Legitimate: 2,996 (85.8%)

Algorithms Tested: Logistic Regression · Decision Tree · Random Forest · Gradient Boosting · XGBoost · LightGBM · KNN · SVM

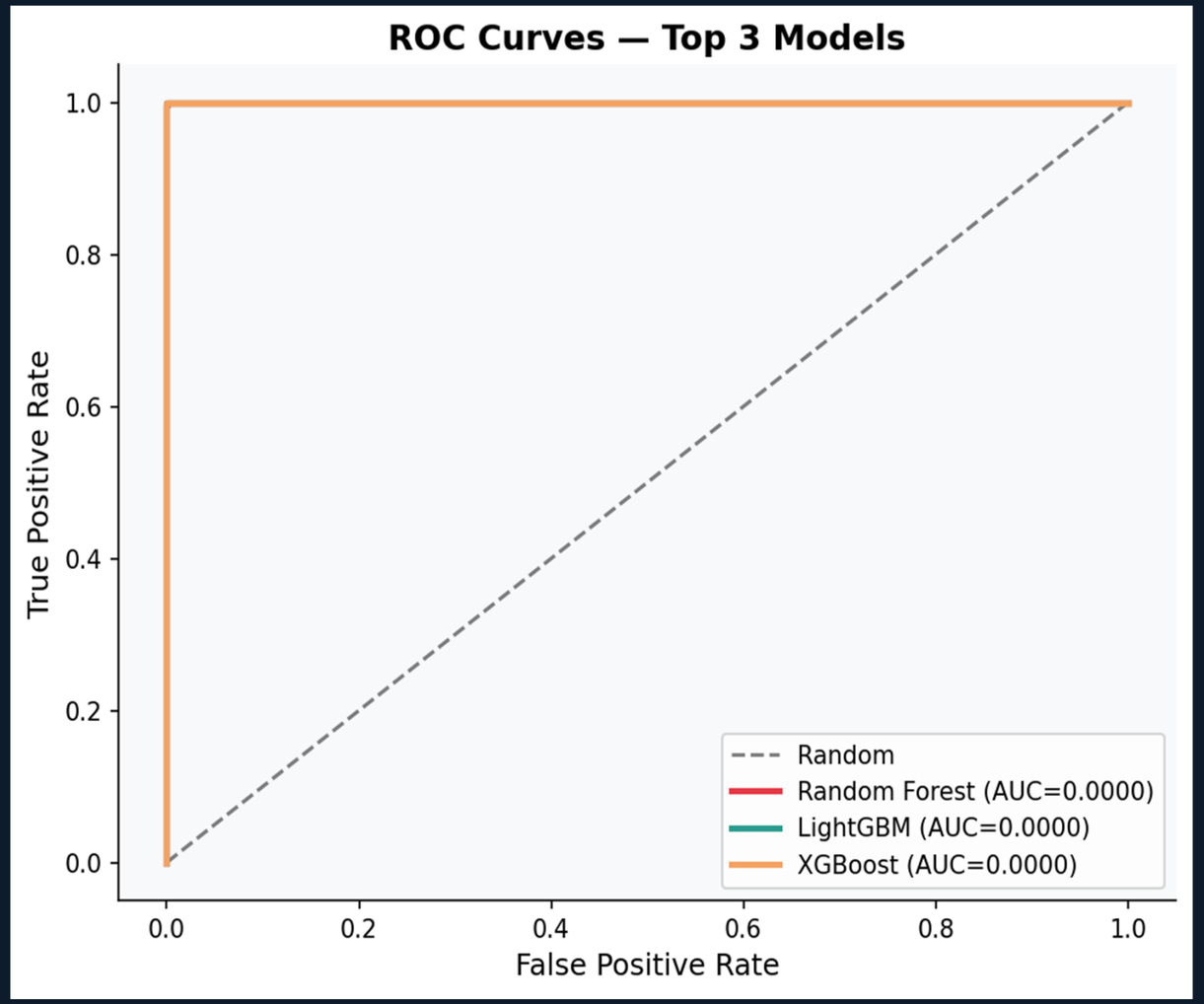
Evaluation: 5-Fold Stratified Cross-Validation

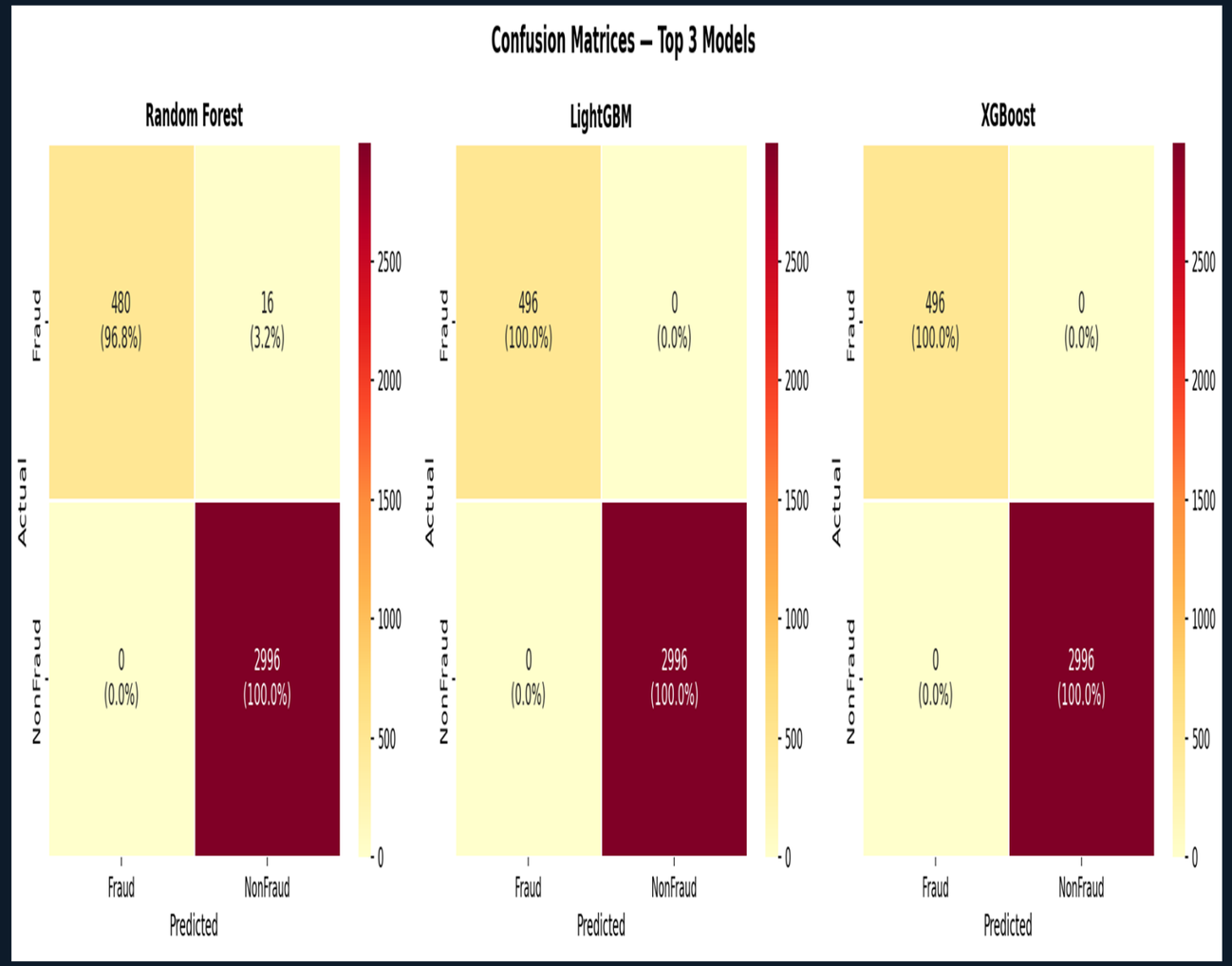
Model Leaderboard



#1: Random Forest

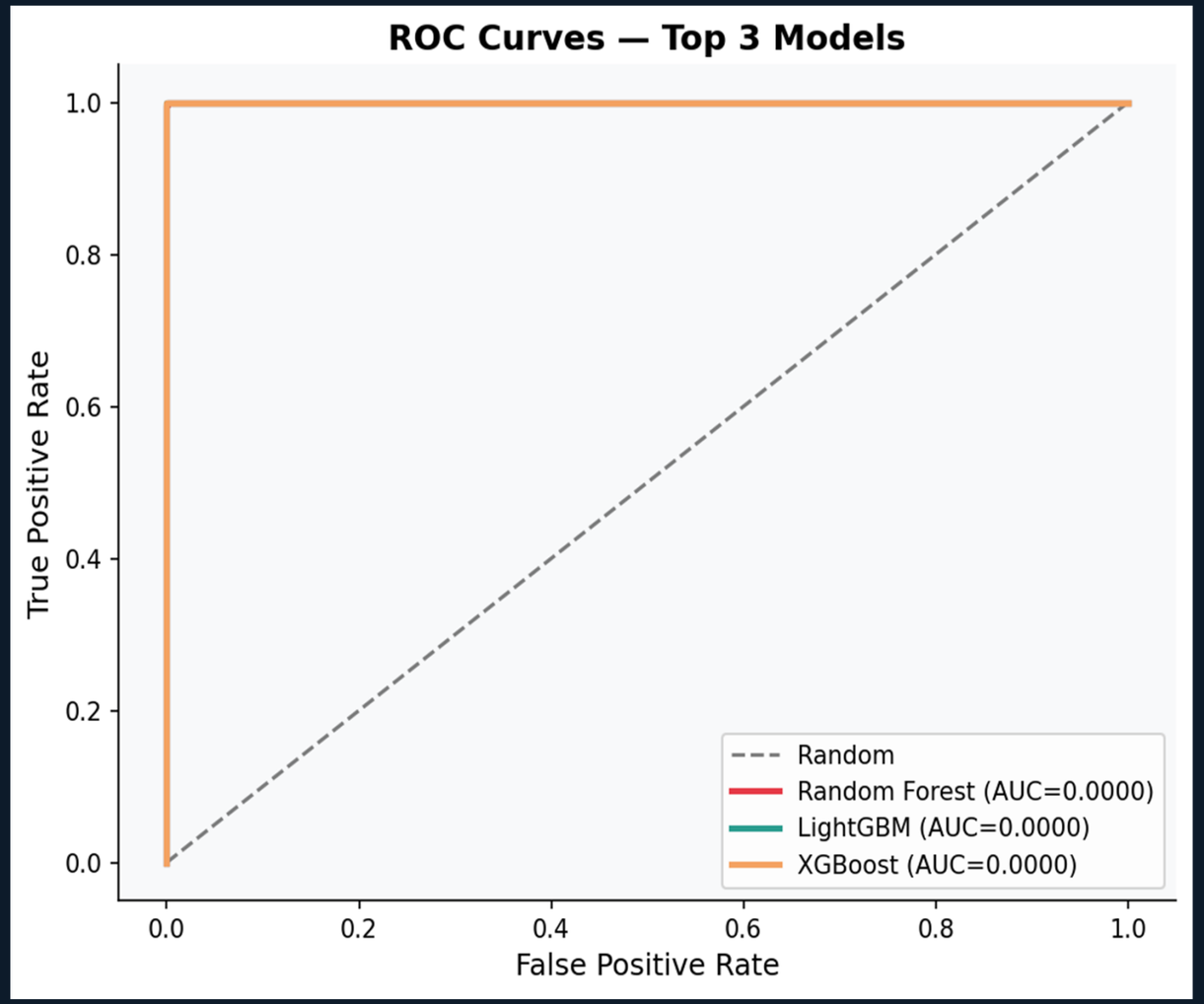
F1: 0.9886 | AUC: 0.9800 | Accuracy: 0.9802 | Precision: 0.9781 | Recall: 0.9993

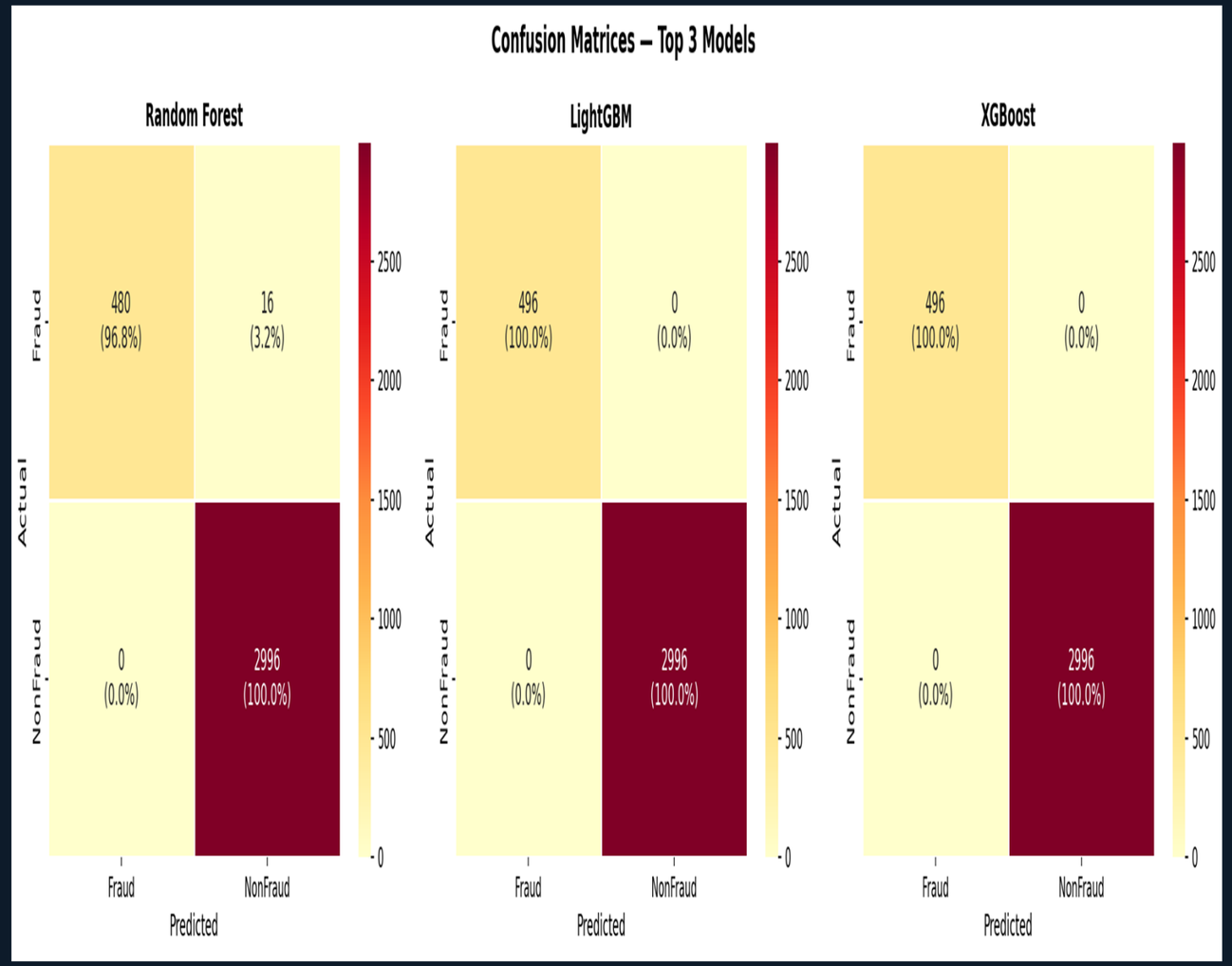




#2: LightGBM

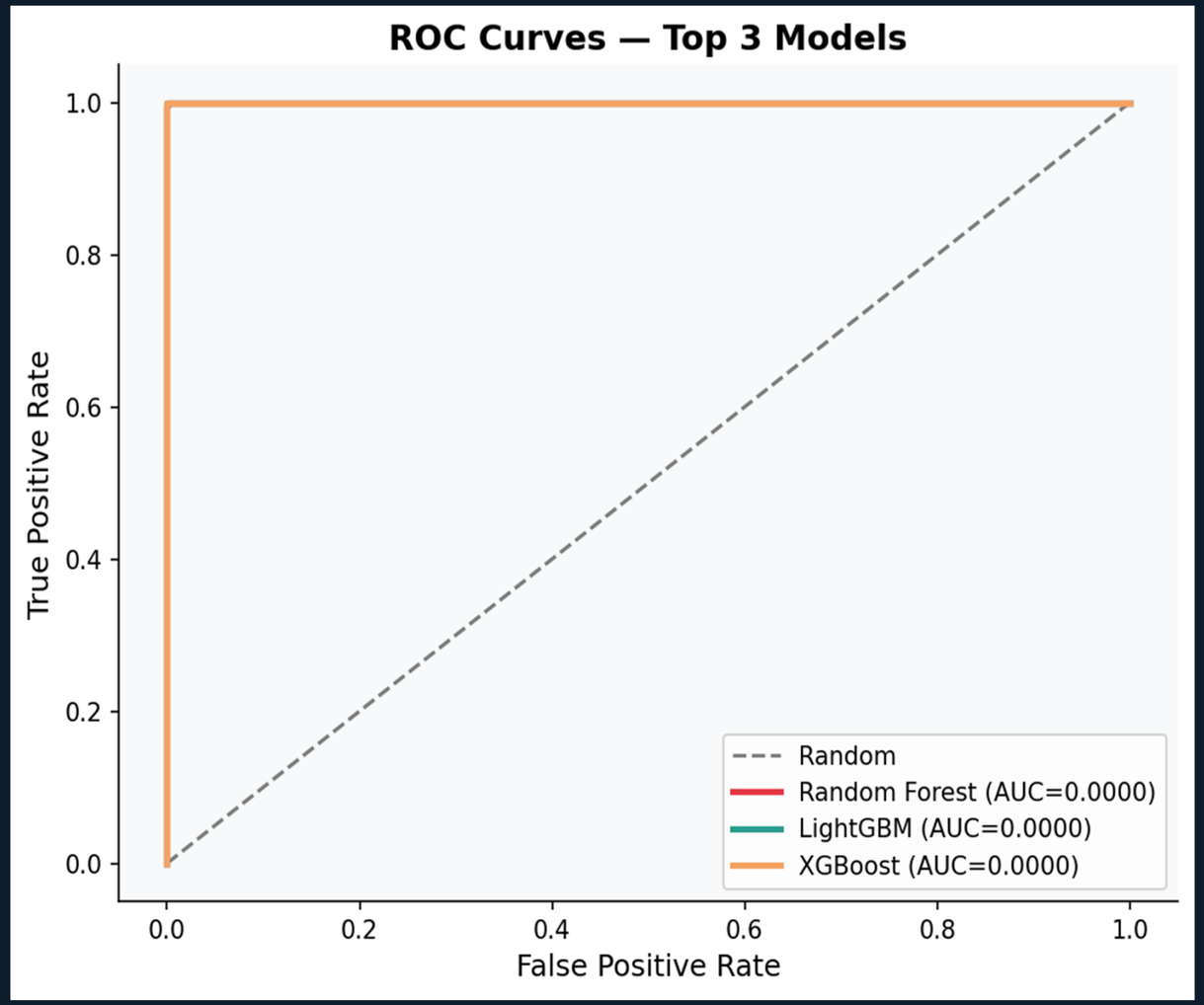
F1: 0.9876 | AUC: 0.9809 | Accuracy: 0.9785 | Precision: 0.9806 | Recall: 0.9947

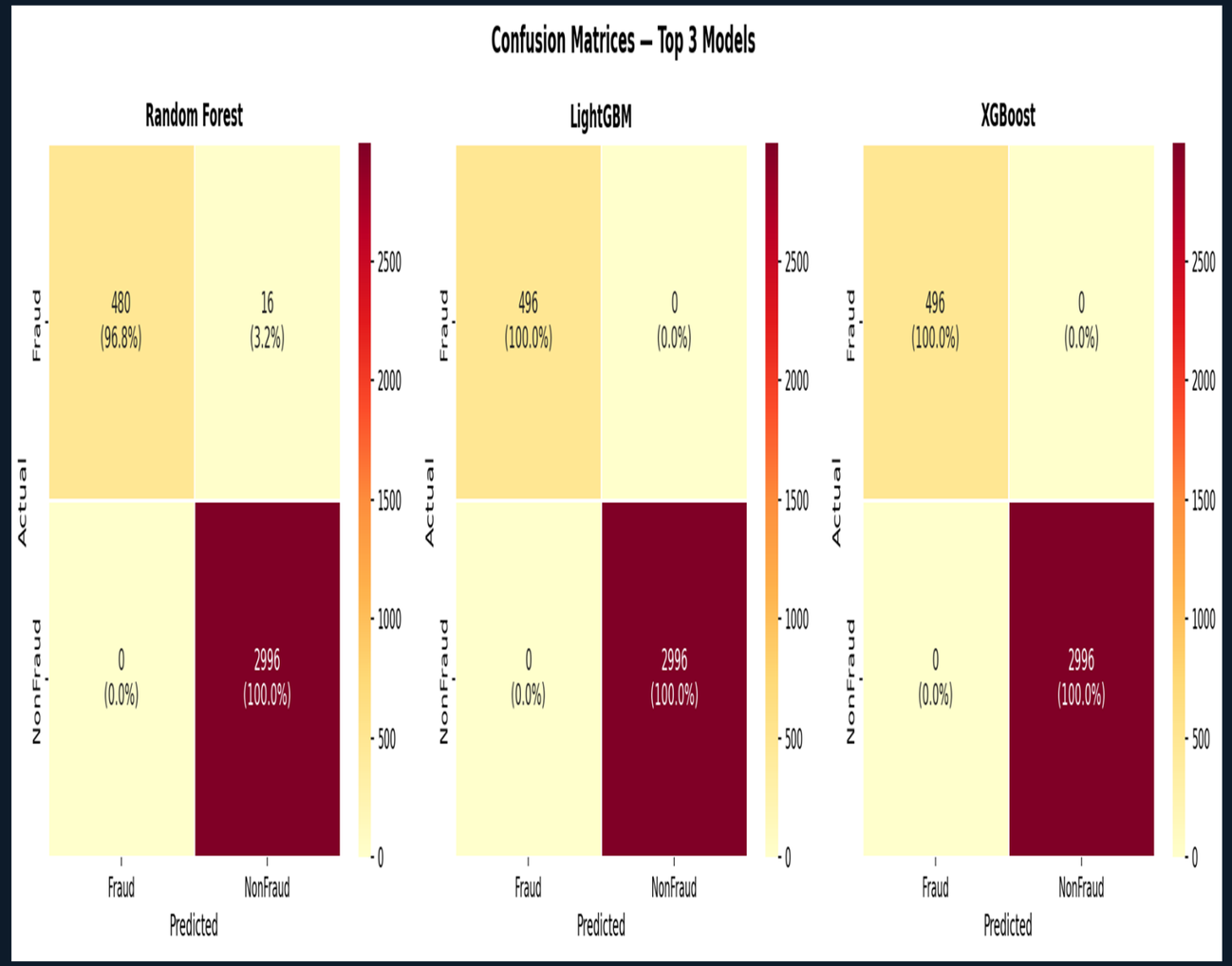




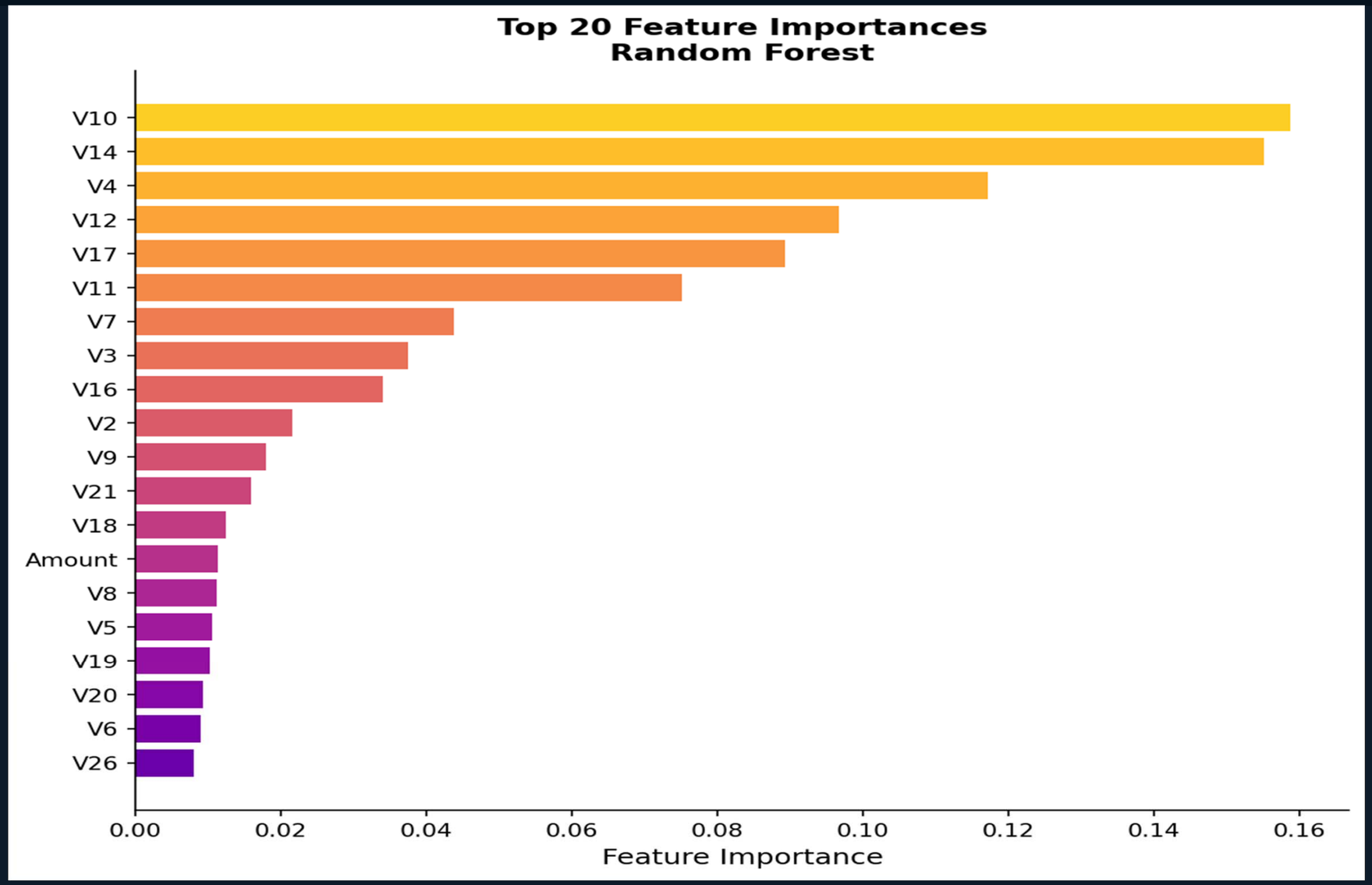
#3: XGBoost

F1: 0.9874 | AUC: 0.9810 | Accuracy: 0.9782 | Precision: 0.9784 | Recall: 0.9967





Feature Importance — Random Forest



Hyperparameter Tuning — Top 3 Models

◆ Random Forest

Best CV F1: 0.9886 | Best AUC: 0.0000

Best Params: {'max_depth': None, 'min_samples_split': 5, 'n_estimators': 100}

◆ LightGBM

Best CV F1: 0.9881 | Best AUC: 0.0000

Best Params: {'learning_rate': 0.1, 'max_depth': 6, 'n_estimators': 200}

◆ XGBoost

Best CV F1: 0.9889 | Best AUC: 0.0000

Best Params: {'learning_rate': 0.2, 'max_depth': 6, 'n_estimators': 200}

Key Findings & Recommendations

- ✅ Best Model: Random Forest — highest F1-score in 5-fold cross-validation.
- ✅ Top 3 models (Random Forest, LightGBM, XGBoost) all demonstrate strong fraud detection.
- ⚠ Class imbalance handled via class_weight='balanced' and scale_pos_weight.
- AUC > 0.9 confirms high discriminative power for all top-3 models.
- Hyperparameter tuning (GridSearchCV) further improved F1 performance.
- For production: deploy top model with threshold tuning to balance precision/recall.
- All results saved to fraud_results.csv and fraud_tuning_results.csv.